

First Demonstration of Novel Combo-ONUs for Cost-Effective Evolution from XGS-PON to 50G-PON

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Abstract: We demonstrate the first 50G/XGS Combo-ONU deployment for PON evolution. It integrates XGS-PON and 50G-PON links. The business modeling and field trial results demonstrate reduced migration cost, operational flexibility, and enhanced system features for operators.

Introduction

Several generations of passive optical networks (PONs) have been deployed using wavelength-based coexistence in a shared optical distribution network (ODN). This includes G-PON, XG(S)-PON, and 50G-PON. In the coexistence architecture, different-generation OLT ports are connected through a coexistence element (CE) [1], while each ONU supports only a single PON technology. Although users can be upgraded by replacing ONUs, operators must deploy new-generation OLT ports across the service area in advance. This leads to significant upfront capital investment, slow return cycles, and unutilized capacity in early stages. It also increases inventory and operational complexity for managing multiple ONU types. A more flexible and demand-driven upgrade mechanism is therefore needed.

Advances in micro-optical integration have enabled cost-effective multi-PON modules [2, 3]. This was adopted on the OLT side as multi-PON module (MPM)-OLT [4]. MPM-OLT has been massively deployed in Asia, representing a very large volume of optical components. The incremental cost of an OLT that supports both XGS-PON and GPON is relatively modest.

In this paper, we extend this concept to the user side and propose the Combo-ONU, which integrates XGS-PON and 50G-PON in a single device. The Combo-ONU enables progressive deployment and on-demand service upgrades. This method reduces front-loaded investment and simplifies migration, with only modest incremental cost compared to single-generation ONUs. In the rest of this paper, we present the design principles, evolutionary deployment strategy, and field trial validation of the 50G/XGS Combo-ONU solution. The results demonstrate that Combo-ONU not only reduces migration cost and operational complexity, but also enhances network

flexibility and service capability.

Combo-ONU for an efficient upgrade to 50G-PON

Figure 1 shows the basic concept of the 50G/XGS Combo-ONU and how it would be used in the introduction of 50G-PON into an existing XGS-PON. The functional block diagram of the Combo-ONU is given in Fig. 1a.

Figure 1b illustrates the proposed PON evolution scheme. When 50G service is first introduced, early adopters are deployed with Combo-ONUs but continue operating on the existing XGS-PON, since the OLT has not yet been upgraded. As user adoption grows, the operator upgrades the OLT to a 50G/XGS MPM-OLT. The deployed Combo-ONUs can then switch to the 50G-PON, freeing XGS resources for remaining users. The 50G/XGS Combo-ONU is slightly more complex than a simplex 50G ONU due to additional optical components, but these parts are low-cost and benefit from optimized micro-optical integration, with only a few extra analog components are required. The four optical links are multiplexed through a WDM, by optimizing the free-space coupling, the total insertion loss is 0.5 dB. In order to meet class C+ 32 dB power budget requirement, a multiple quantum well (MQW) 50G DML was employed [5], the ridge waveguide structure effectively enhanced the laser slope efficiency, as shown in Fig. 1c. Its max output power coupled into fiber exceeds 11 dBm, as shown in Fig. 1d. After the pre-equalizer, the ER and TDEC of 50Gb/s signal were measured to be 6.5 dB and 1.4 dB, respectively. In practice, an XGS/GPON Combo-ONU costs only about 10% more than a simplex XGS ONU. With large-scale production, 50G/XGS Combo-ONUs are expected to follow a similar cost trend.

Figure 2 compares PON upgrade strategies ver-

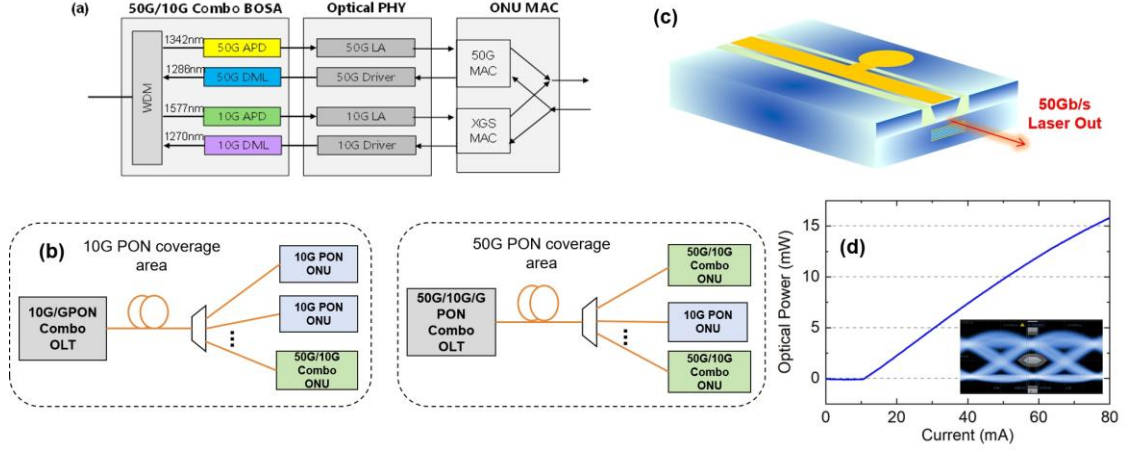
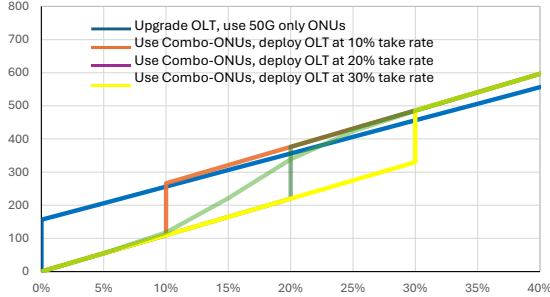


Fig. 1: Structure of 50G/XGS Combo-ONU and how it is used during upgradation from XGS-PON to 50G-PON, (a) Combo-ONU function blocks, (b) XGS to 50G-PON evolution with Combo-ONUs, (c) structure of high-power MQW DML, (d) output power test results of 50G DML.

50G take rate. In the conventional coexistence model (*i.e.*, the blue curve), 50G OLTs are deployed upfront, resulting in high initial capital expenditure before customer adoption. In contrast, the Combo-ONU model deploys Combo-ONUs first and defers OLT upgrades until adoption reaches certain value (*e.g.*, 20%), significantly reducing early investment and financial risk. This aligns major OLT spending with revenue growth. It enables gradual and port-level migration from XGS to 50G PON without disruptive upgrades.



N : total subscribers under an OLT port
 $R \in [0,1]$: 50G-PON take rate
 C_{OLT50} : cost of one 50G-PON OLT port
 C_{ONU50} : cost of a 50G-PON-only ONU
 $C_{ONUCombo}$: cost of a Combo-ONU
 C_{OLTMPM} : cost of MPM-OLT port
 r_t : take rate threshold triggering OLT upgrade

$$C_{Conventional} = C_{OLT50} + rN \cdot C_{ONU50} \quad (1)$$

$$C_{Combo} = \begin{cases} rN \cdot C_{ONUCombo}, & \text{when } r < r_t \\ r_t N \cdot C_{ONUCombo} + (r - r_t) N \cdot C_{ONU50} + C_{OLTMPM}, & \text{when } r \geq r_t \end{cases} \quad (2)$$

Fig. 2: Upgrade costs comparison with evaluation model.

Field trial and test results

A field trial of the 50G/XGS Combo-ONU solution was conducted in Suzhou, Jiangsu Province. The original deployment in the trial area was based on XGS-PON, with an OLT serving ONUs through a

1:64 splitter. As traffic demand increased additional capacity became necessary, these users' ONUs were first replaced by 50G/XGS Combo-ONUs. When more than 12 high-bandwidth users (~20%) emerged in this area, the OLT was replaced with a 50G/XGS MPM combo OLT. In this way, high-demand users were upgraded to access 50G capacity, while remaining users continued operating on XGS-PON. The field trial configuration and main performances are illustrated in Fig. 3.



	Original PON	Field trial upgrade
Downstream rate	10G	10G and 50G
Upstream rate	10G	10G and 50G
ODN	1:64 split ratio, ~5 km trunk fiber, <= 100 m feeder fiber	
Users	~50 residential	~50 residential, ~10 business
OLT MPM take rate	~20%	
ONUs	All XGS ONUs	~30 Combo-ONUs, ~30 XGS ONUs
PON Link budget	32 dB, C+ class (XGS)	32 dB, C+ class (XGS) 32 dB, C+ class (50G)

Fig. 3: Field trial configuration and map for Combo-ONU in China Mobile Jiangsu branch.

With the introduction of Combo-ONUs that have dual optical interfaces, a new class of system capabilities become possible. One capability is fault protection. Traditional PON protection schemes typically require duplicate optical interfaces with

almost double ONU cost. With Combo-ONU, the protecting interface is of the previous generation (i.e., XGS), and the cost is greatly reduced because it has enjoyed the economies of scale over its life cycle. As shown in Fig. 4, when a fault occurs on the 50G channel, the failure is detected and reported, triggering a protection switching command. The ONU then switches traffic to the XGS channel through the shared WDM interface, restoring service continuity. Although the bandwidth capabilities are reduced, it ensures uninterrupted connectivity. In a shared bandwidth system like PON, temporary bandwidth degradation is generally preferable over no service, and most applications can adapt to the network. As reported in the field trial validation in Fig. 4, when a fault was introduced on the 50G channel, traffic was switched to the XGS channel in less than 50 ms. The packet loss rate remained below 0.01%, and no ONU re-registration was required.

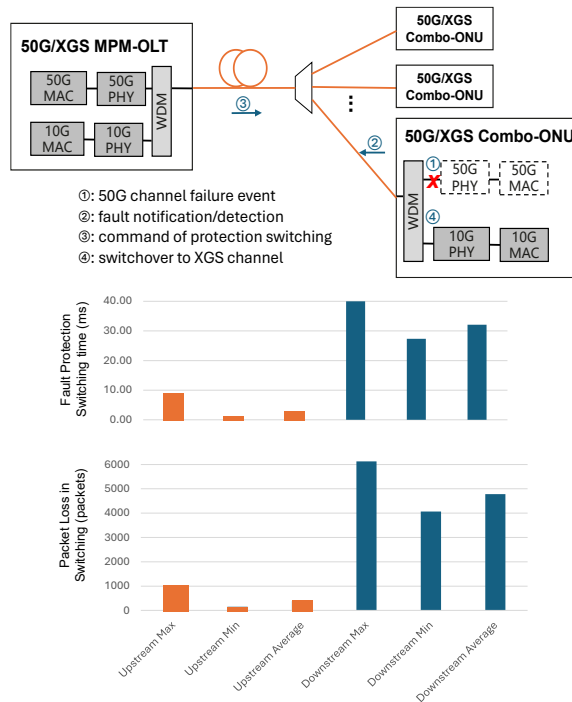


Fig. 4: Combo-ONU for fault protection and test results.

Another capability tested was for low latency and jitter services. One of the most fundamental challenges of PON is the necessity of ranging, which requires the OLT to open a ranging window that is as long as the roundtrip time (typically $\sim 250 \mu\text{s}$). Ranging windows temporarily interrupt upstream transmission and introduce latency and jitter. In the Combo-ONU architecture, the legacy XGS interface can be utilized for ONU activation and ranging, while the 50G interface remains dedicated to low latency services. As shown in Fig. 5, by separating control-plane operations from high-speed data traffic, the system avoids ranging-induced interruptions on the 50G channel. This approach enables more deterministic

performance, providing reduced latency and jitter for delay-critical services without requiring complex out-of-band ranging or additional management mechanisms. The field trial confirmed the low-latency capability of the 50G/XGS Combo-ONU architecture. Measured PON segment latency remained below $150 \mu\text{s}$, with a jitter controlled to be less than $100 \mu\text{s}$.

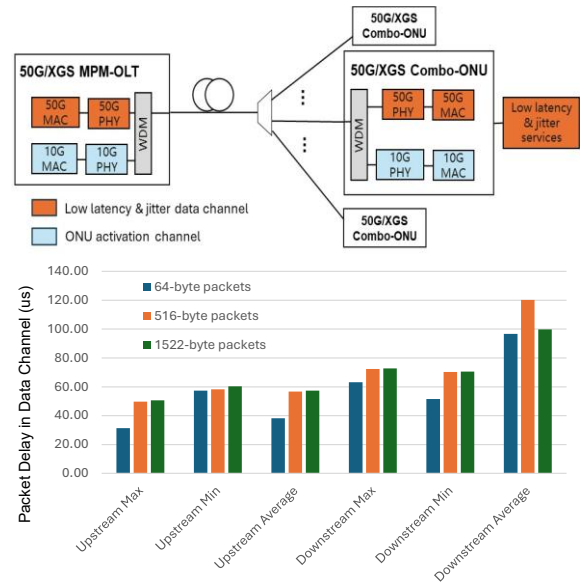


Fig. 5: Combo-ONU for low latency and jitter services with test results.

Conclusions

This paper presents the first field trial of a Combo-ONU solution enabling smooth evolution from XGS-PON to 50G-PON. By integrating both interfaces within a single ONU, the architecture supports coexistence and migration while reducing upgrading costs and improving operational flexibility. A 5 km, 1:64 field trial validated XGS to 50G upgrade with a 32 dB C+ loss budget. Fault protection tests demonstrated switchover $< 50 \text{ ms}$ with negligible packet loss, and latency tests achieved $< 150 \mu\text{s}$ delay with $< 100 \mu\text{s}$ jitter. This field trial provides a practical evidence to support operators in making informed decisions for large-scale XG(S)-PON evolution towards 50G-PON.

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